

1) Dataset: [7, 14, 21, 28, 35, 42, 49, 56]

key to Search: 49

Interpretation

The dataset is already Sorted. we compare Linear & Binary Search to find element 49 & determine which algorithm is efficient

Comparison	Element	Result
	7	Not Found
1	14	"
2	21	"
3	28	"
4	35	"
5	42	"
6	49	Found
7		

Total Comparisons: 7

1) middle = 28 $\rightarrow$ 49 > 28 $\rightarrow$ search right half :

2) middle = 42 $\rightarrow$ 49 > 42 $\rightarrow$ Search right half

3) middle = 49 $\rightarrow$ Found

Total Comparison: 3

Time Complexity

- Linear Search

- Best case: $O(1)$

- Avg case: $O(n)$

- Worst case: $O(n)$

- Binary Search
- Best Case : $O(1)$
- Avg Case : $O(\log n)$
- Worst Case : $O(\log n)$

Comparison :

Feature	Linear Search	Binary Search
Comparisons	7	3
Best for	Small / unsorted data	Sorted data
Time Complexity	$O(n)$	$O(\log n)$

Results :

Binary Search is more efficient because it finds the element in only 3 comparisons while Linear Search requires 7 comparisons. Binary Search is preferred for sorted datasets.

2)

Dataset: [6, 12, 18, 24, 30, 36, 42, 48]

Key to Search: 36

Interpretation

The dataset is Sorted. The objective is to Compare Linear Search & Binary Search while Searching for 36

Linear Search

Comparison

Element

Result

1

6

Not Found

2

12

Not Found

3

18

"

4

24

"

5

30

"

6

36

Found

Total Comparisons: 6

Binary Search

1. Middle = 24 $\rightarrow$ 36 > 24 $\rightarrow$ Search right half

2. Middle = 36 $\rightarrow$ Found

Total Comparison: 2

Time Complexity:

- Linear Search
 - Best Case: $O(1)$
 - Avg Case: $O(n)$
 - Worst Case: $O(n)$
- Binary Search
 - Best Case: $O(1)$
 - Avg Case: $O(\log n)$
 - Worst Case: $O(\log n)$

Comparison:

Feature	Linear Search	Binary Search
Comparisons	6	2
Best for	Small/unsorted data	Sorted Data
Time Complexity	$O(n)$	$O(\log n)$

Result:

Binary Search performs better because it requires only 2 comparison, where Linear Search requires 6 comparison. Since Binary Search repeatedly divides the search space into two halves, it is faster, more scalable & the preferred method for searching in sorted data.